

## LETTERS TO THE EDITOR

**“Meta-epidemiological study” is a study in which the unit of analysis is a study, not a patient; response to Puljak et al.**



Dear Editor,

We welcome Prof. Puljak’s comment and related endeavors to clarify and define the currently confusing nomenclatures around “meta-epidemiology” and related terms such as “meta-research” and “research-on-research” [1,2]. We agree that the concept of “meta-epidemiological” studies includes studies that evaluate the impact of trial characteristics on effect sizes [3]. However, we do not agree with limiting the concept of this term only to such studies for several reasons.

The definitions of technical terms can be to some degree arbitrary, i.e., we can define whatever and whichever way we agree with as a scientific community. However, such definitions should be conceptually coherent (i.e., the intention of the term must be clearly defined) and transparent about what it includes and what it excludes (i.e., the extension of the term must be easily understood); furthermore, it will be better if the definition is historically consistent.

Historically, the first usage of the term “metaepidemiology” appeared in Naylor’s editorial in The British Medical Journal, in which he called a narrative account of systematic reviewers’ efforts to seek unpublished data from the original authors “the new meta-epidemiology of randomized trials” [4]. The second instance of the term appeared in Sterne’s influential paper which explained statistical methods for assessing the influence of study characteristics on treatment effects [3]; these authors apparently

characterized such studies as one type of meta-epidemiological research but not necessarily the only one.

Furthermore, if researchers use the narrow definition of “metaepidemiological”, as suggested by Prof. Puljak, we agree that the understanding of the term itself in the scientific community will be facilitated. However, then, the use of the alternative but heterogeneous words such as “meta-research”, “methodological studies”, and “research-on-research studies” will proliferate. This will not improve the accessibility of the medical literature.

The greatest common defining concept of “metaepidemiological studies” would be any studies in which the unit of analysis is a study, not a patient [5]. This definition is clear and consistent with the fact that “meta” was the Greek word for “after” or “beyond” [6]. In other words, metaepidemiological studies are studies conducted after epidemiological studies. Without getting too locked in the concept at a temporal point, we are willing to deepen our research in this field.

Yuki Kataoka

Department of Internal Medicine

Kyoto Min-iren Asukai Hospital

Kyoto, Japan

Scientific Research Works Peer Support Group (SRWS-PSG)

Osaka, Japan

Section of Clinical Epidemiology

Department of Community Medicine

Kyoto University Graduate School of Medicine

Kyoto, Japan

Department of Healthcare Epidemiology

Kyoto University Graduate School of Medicine / School

of Public Health

Kyoto, Japan

Masahiro Banno

Department of Psychiatry

Seichiryo Hospital

Nagoya, Japan

Department of Psychiatry

Nagoya University Graduate School of Medicine

Nagoya, Japan

Scientific Research WorkS Peer Support Group (SRWS-PSG)

Osaka, Japan

Yasushi Tsujimoto

Oku Medical Clinic

Osaka, Japan

DOI of original article: <https://doi.org/10.1016/j.jclinepi.2022.12.003>.

Author contributions: Yuki Kataoka had full access to all the data in the study and took responsibility for the integrity of the data and the accuracy of the data analysis. Drafting of the manuscript: YK. Critical revision of the manuscript for important intellectual content: Masahiro Banno, Yasushi Tsujimoto, and Toshi A. Furukawa. All authors gave final approval of the version to be published and agreed to be accountable for all aspects of this work.

Conflict of interest: Yuki Kataoka: none known. Masahiro Banno: none known. Yasushi Tsujimoto: none known. Toshi A. Furukawa: TAF reports personal fees from DT Axis, Kyoto University Original, Merck Sharp & Dohme and Sony Group Corporation and a grant from Shionogi, outside the submitted work; In addition, TAF has patents 2020-548587 and 2022-082495 pending, and intellectual properties for Kokoro-app licensed to Mitsubishi-Tanabe.

Funding: none.

Department of Health Promotion and Human Behavior  
 Kyoto University Graduate School of Medicine/School of  
 Public Health  
 Kyoto University, Kyoto, Japan  
 Scientific Research WorkS Peer Support Group (SRWS-PSG)  
 Osaka, Japan

Toshi A. Furukawa  
 Department of Health Promotion and Human Behavior  
 Kyoto University Graduate School of Medicine/School of  
 Public Health  
 Kyoto, Japan

\*Corresponding author: Department of Health Promotion and  
 Human Behavior, Kyoto University Graduate School of Medicine/  
 School of Public Health, Kyoto, Japan, Tel.: +81 75 753 9491;  
 fax: +81 75 753 4641.

E-mail address: [furukawa@kuhp.kyoto-u.ac.jp](mailto:furukawa@kuhp.kyoto-u.ac.jp) (T.A. Furukawa)

## References

- [1] Puljak L. Caution is needed when describing a study design as meta-epidemiological. *J Clin Epidemiol* 2022;152:326–7.
- [2] Puljak L, Makaric ZL, Buljan I, Pieper D. What is a meta-epidemiological study? Analysis of published literature indicated heterogeneous study designs and definitions. *J Comp Eff Res* 2020;9(7):497–508.
- [3] Sterne JAC, Jüni P, Schulz KF, Altman DG, Bartlett C, Egger M. Statistical methods for assessing the influence of study characteristics on treatment effects in “meta-epidemiological” research. *Stat Med* 2002;21:1513–24.
- [4] Naylor CD. Meta-analysis and the meta-epidemiology of clinical research. *BMJ* 1997;315:617–9.
- [5] Murad MH, Wang Z. Guidelines for reporting meta-epidemiological methodology research. *Evid Based Med* 2017;22(4):139–42.
- [6] That’s so meta [Internet]. 2017. Merriam-Webster. Available at: <https://www.merriam-webster.com/words-at-play/meta-adjective-self-referential>. Accessed December 17, 2022.

<https://doi.org/10.1016/j.jclinepi.2022.12.002>

## Challenges with defining a meta-epidemiological study as “study in which the unit of analysis is a study, not a patient”: author’s reply



Dear Editor:

I appreciate Kataoka et al. [1] addressing my suggestion that caution is needed when labeling a study as “meta-epidemiological” [2].

My intention was not to impose a narrow definition of a “meta-epidemiological study” in line with its origins [3]. Instead, I intended to point out that many authors diverge

DOI of original article: <https://doi.org/10.1016/j.jclinepi.2022.12.002>.

Livia Puljak is the sole author of this manuscript. She designed and wrote the article.

Livia Puljak has published research on this topic before. She has no conflict of interest to declare related to this topic.

This article is not associated with any extramural funding.

Raw data were not generated.

from the definition from 2002 [4] and, consequently, the meaning of the descriptor “meta-epidemiological study” is nowadays unclear, without consensus in the research community.

## Going back to the origins

There are multiple challenges with the definition proposed by Kataoka et al. [quote]: “Meta-epidemiological study is a study in which the unit of analysis is a study, not a patient” [1]. Their definition is inspired by the article of Murad and Wang, published in 2017 [5], which indicated [quote]: “Meta-epidemiological studies adopt a systematic review or meta-analysis approach to examine the impact of certain characteristics of clinical studies on the observed effect and provide empirical evidence for hypothesised associations. The unit of analysis in meta-epidemiological studies is a study, not a patient” [5].

Thus, Murad and Wang [5] did not define a meta-epidemiological study simply as a study where the unit of analysis is a study. To me, their definition sounds very similar to the definition by Sterne et al. from 2002 [3], indicating that those are studies “in which the influence of trial characteristics such as measures of study quality on treatment effect estimates is explored.”

## Systematic reviews also use “a study” as a unit of analysis

As systematic reviews also use “a study” as a unit of analysis, the proposed broad definition of Kataoka et al. [1] would encompass systematic reviews as well. It would be very important to use different descriptors for evidence syntheses and studies analyzing research methods [6]. In this respect, the proposed definition would be too broad.

## What is “a study”?

Furthermore, it is debatable what is a “study,” as this term could be very restrictive. Thus, the proposed broad definition may not be broad enough. For example, methodological studies can also use human participants [7]. Studies labeled as “meta-epidemiological” in the title have analyzed units of analysis that may not be considered “a study,” such as guidelines [8], protocols [9], PROSPERO records [10], and journals’ author instructions [11].

One study, labeled as meta-epidemiological, took Cochrane reviews and repeated literature searches using multiple abbreviated approaches. They conducted 840 abbreviated literature searches to analyze their effects on evidence syntheses conclusions [12]. Thus, it is highly questionable whether this could be considered research where the unit of analysis is “a study.” There were many other study designs in the field of research methodology where it would be difficult to say that the unit of analysis was a study.

In conclusion, the definition of a “meta-epidemiological study” could certainly evolve from its first description published 20 years ago [3]. Evolution involves adaptation over